

**VELOX** PEPTIDES



# The Velox Researcher's Handbook

A plain-English reference to research peptides,  
handling and documentation.

• **≥99% HPLC**

• **Janoshik tested**

• **CoA per batch**

**2026 Edition**  
[veloxpeps.com](https://veloxpeps.com)

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Supplied strictly as research reagents for in vitro use only. Not for human or  
veterinary consumption, injection, dosing, diagnosis or treatment.

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Holywood, Northern Ireland

# Contents

## Part 1 — Working with research peptides

|                                       |   |
|---------------------------------------|---|
| What research peptides are            | 3 |
| Reconstitution & concentration        | 4 |
| Storage & handling                    | 5 |
| How to read a Certificate of Analysis | 6 |

## Part 2 — Compound reference

|                      |    |
|----------------------|----|
| • <b>Cognitive</b>   | 7  |
| Semax                | 8  |
| Selank               | 9  |
| DSIP                 | 10 |
| • <b>Metabolic</b>   | 11 |
| Retatrutide          | 12 |
| MOTS-c               | 13 |
| NAD+                 | 14 |
| • <b>Recovery</b>    | 15 |
| BPC-157              | 16 |
| TB-500               | 17 |
| KPV                  | 18 |
| GHK-Cu               | 19 |
| • <b>Growth</b>      | 20 |
| CJC-1295             | 21 |
| Tesamorelin          | 22 |
| • <b>Antioxidant</b> | 23 |
| Glutathione          | 24 |
| Melanotan II         | 25 |

## Glossary 26

## Why Velox · How to order 27

# Working with research peptides

## What research peptides are

Peptides are short chains of amino acids - the same building blocks that make up proteins. Research peptides are supplied as research reagents (chemicals used in laboratory work), intended strictly for in vitro use: test-tube and cell-culture experiments. They are not medicines, are not approved by the MHRA or FDA, and are never for human or veterinary use, consumption, injection, dosing, diagnosis or treatment.

Because peptides are sensitive molecules, the quality of a research result depends as much on correct handling and documentation as on the compound itself. The pages that follow cover the three things every researcher should get right: reconstituting the powder accurately, storing it correctly, and reading the Certificate of Analysis that proves what is in the vial.

### Three pillars of good practice

Reconstitute accurately · Store cold and stable · Verify every batch on its CoA

## PART 1

# Reconstitution & concentration

## Mixing the powder back into a liquid

Peptides are supplied lyophilised (freeze-dried). Before use in research they are reconstituted - mixed back into a liquid, almost always bacteriostatic water (sterile water containing 0.9% benzyl alcohol, which limits bacterial growth so a vial can be drawn from over several uses).

### CONCENTRATION FORMULA

$$\text{peptide (mg)} \div \text{water (mL)} = \text{concentration (mg/mL)}$$

## Worked example

A 10 mg vial reconstituted with 2 mL of bacteriostatic water gives  $10 \div 2 = 5$  mg/mL. Drawing 0.2 mL from that vial therefore contains 1 mg of peptide. Velox provides an online reconstitution calculator at [veloxpeps.com/tools/reconstitution-calculator/](https://veloxpeps.com/tools/reconstitution-calculator/).

## Quick-reference: concentration (mg/mL)

| Vial size | +1 mL | +2 mL | +3 mL | +5 mL |
|-----------|-------|-------|-------|-------|
| 5 mg      | 5.0   | 2.5   | 1.7   | 1.0   |
| 10 mg     | 10.0  | 5.0   | 3.3   | 2.0   |
| 15 mg     | 15.0  | 7.5   | 5.0   | 3.0   |
| 20 mg     | 20.0  | 10.0  | 6.7   | 4.0   |

# Storage & handling

## Keeping peptides stable

Lyophilised peptide is stable for long periods when kept frozen and dry. Once reconstituted, a peptide is in solution and far more fragile, so it is kept refrigerated and used within a shorter window. The single biggest avoidable error is repeated freeze-thaw cycling, which degrades peptides quickly; aliquot into smaller portions instead. Always use sterile technique, and discard any solution that turns cloudy, changes colour or shows particles.

## Storage quick-reference

| State                         | Temperature        | Notes                                    |
|-------------------------------|--------------------|------------------------------------------|
| <b>Lyophilised powder</b>     | −20°C (or below)   | Stable long-term; keep dry and sealed    |
| <b>Reconstituted solution</b> | 2–8°C (fridge)     | Use within a short window; do not freeze |
| <b>In use</b>                 | Minimise warm time | Return to fridge promptly after drawing  |

## Signs of degradation

Cloudiness, discolouration, visible particles or an off smell all indicate a solution should be discarded. Lyophilised powder that has clumped or changed colour should likewise not be used. Good labelling - compound, concentration, date reconstituted - keeps a research record clean and traceable.

# How to read a Certificate of Analysis

## What a CoA proves

A Certificate of Analysis (CoA) is the lab report for a specific batch. It is the document that tells you what is actually in the vial. Velox supplies a batch CoA on request and uses Janoshik, an independent third-party testing laboratory, so the result is not self-reported.

- **HPLC purity %**

High-performance liquid chromatography separates the sample and measures how much is the target peptide versus impurities. Velox compounds are verified at  $\geq 99\%$ .

- **Mass spec / molecular weight**

Mass spectrometry confirms the measured molecular weight matches the expected weight for that exact peptide - proof of identity, not just purity.

- **Appearance**

A short physical description (e.g. white lyophilised powder) confirming the sample looks as it should.

- **PASS**

An overall result indicating the batch met its specification across the tests performed.

## Why third-party testing matters

An independent lab (Janoshik) has no incentive to inflate a result - so the number you read is one you can trust

CATEGORY

# Cognitive

Neuropeptides studied for brain-growth proteins, calming signals and sleep.

IN THIS SECTION

- **Semax**
- **Selank**
- **DSIP**

**VELOX** PEPTIDES

# Semax

**CAS****80714-61-0****CLASS****ACTH(4-7) analogue ·  
Met-Glu-His-Phe-Pro-Gly-Pro****HPLC****≥99.0%**

## What it is

Semax is a lab-made heptapeptide (a chain of seven amino acids) copied from a small piece of the natural hormone ACTH. It is a neuropeptide - a peptide that acts on nerve cells. A Pro-Gly-Pro tail slows enzyme breakdown, so it lasts far longer than the plain fragment while keeping the nerve-signalling part without the stress-hormone part.

## What's been studied

- Linked to raised BDNF and NGF (proteins that help nerve cells grow and survive), with changes at the TrkB receptor in rodent brain.
- Gene studies show region-specific, multidirectional regulation of neurotrophin activity rather than one uniform change.
- In rodent stroke (ischaemia) and oxidative-stress models, associated with switching on nerve-growth genes and dialling down inflammation and cell-death genes.
- Studied for effects on attention, learning and the dopamine and serotonin systems (chemical messengers for mood and focus).

**SUPPLIED AS****Lyophilised powder · ≥99.0% HPLC · batch CoA on request.**



# Selank

**CAS****129954-34-3****CLASS****Tuftsins analogue ·  
Thr-Lys-Pro-Arg-Pro-Gly-Pro****HPLC****≥98.9%**

## What it is

Selank is a lab-made heptapeptide derived from the natural immune peptide tuftsins, with a Pro-Gly-Pro tail added for stability. It is a neuropeptide studied for anxiety-like behaviour in animal models and for its action on the GABA system, the brain's main calming signal.

## What's been studied

- Modulates the GABA-A receptor indirectly (allosterically) - not at the benzodiazepine site - which researchers link to a calming effect without heavy sedation.
- In rodent models, associated with raised BDNF in the hippocampus and with serotonin signalling (memory and mood pathways).
- Because it derives from tuftsins, studied for slowing enkephalin breakdown and for effects on the immune system and cytokines.
- Anxiolytic effects in elevated-plus-maze models reported comparable to diazepam but without motor impairment.

**SUPPLIED AS****Lyophilised powder · ≥98.9% HPLC · batch CoA on request.**

# DSIP

**CAS****62568-57-4****CLASS****Endogenous nonapeptide · 9 amino acids****HPLC****≥98.7%**

## What it is

DSIP (Delta Sleep-Inducing Peptide) is a naturally occurring nonapeptide (nine amino acids), first found in the blood of sleeping rabbits. Its name comes from the slow delta brain waves of deep sleep. Even after decades of study, its exact receptor and mechanism remain unresolved - part of why it is still researched.

## What's been studied

- Best known for links to slow-wave (delta) sleep - its deepest stage - in rat and rabbit studies, alongside more REM sleep.
- Blocking an animal's own DSIP with an antiserum reduced the usual deep-sleep rebound after sleep loss, suggesting a role in natural sleep control.
- Studied for links to sleep-time hormone release, including growth hormone, via a somatostatin/dopamine pathway.
- Also examined for calming the stress response, antioxidant activity and keeping internal systems in balance.

**SUPPLIED AS****Lyophilised powder · ≥98.7% HPLC · batch CoA on request.**

CATEGORY

# Metabolic

Compounds studied for how cells and the body handle energy, fuel and fat.

IN THIS SECTION

- **Retatrutide**
- **MOTS-c**
- **NAD+**

VELOX **PEPTIDES**

# Retatrutide

**CODE****LY3437943 (Eli Lilly)****CLASS****Triple GIP / GLP-1 / glucagon agonist****HPLC****≥99%**

## What it is

Retatrutide is a lab-made peptide built to switch on three receptors (cell-surface “locks”) at once: GLP-1, GIP and glucagon. An agonist is a molecule that turns a receptor on. Each receptor helps control metabolism - how the body handles energy, blood sugar and fat - so hitting all three works in a more complex way than single- or dual-receptor compounds.

## What's been studied

- GLP-1 pathway: linked to glucose-dependent insulin release, slower stomach emptying and appetite signals in the hypothalamus.
- GIP pathway: linked to insulin release and to changes in fat tissue, including fat storage and breakdown (lipolysis) in mice.
- Glucagon pathway: acts mainly on liver cells to control stored-sugar release, with data pointing to more heat production (thermogenesis).
- Preclinical mouse and primate work reported activity at all three receptors and a half-life supporting once-weekly dosing.

**SUPPLIED AS****Lyophilised powder · ≥99% HPLC (third-party) · batch CoA on request.**

# MOTS-c

**CLASS****Mitochondrial-derived peptide  
(16 aa)****ORIGIN****Encoded in mitochondrial 12S  
rRNA****HPLC****≥99%**

## What it is

MOTS-c is a 16-amino-acid peptide that is unusual because its instructions live inside the mitochondria - the cell's tiny power plants - rather than the nucleus. It belongs to a small group called mitochondrial-derived peptides (MDPs) that act as messengers, and is studied mainly as an activator of the AMPK energy sensor.

## What's been studied

- Switches on AMPK (the cell's "low-fuel light") indirectly, by interfering with the folate cycle so the helper molecule AICAR builds up - mainly in skeletal muscle.
- Under stress, moves from the mitochondria into the nucleus and helps switch on defence and antioxidant genes - a rare mitochondrial-to-nuclear signal.
- In rodent studies, linked to better insulin response despite age or a high-fat diet, and to less diet-induced weight gain.
- Described as an exercise-responsive peptide tied to fitness and age-related decline.

**SUPPLIED AS****Lyophilised powder · ≥99% HPLC · batch CoA on request.**

# NAD<sup>+</sup>

**CAS****53-84-9****CLASS****Pyridine dinucleotide  
coenzyme****HPLC****≥99.5%**

## What it is

NAD<sup>+</sup> (nicotinamide adenine dinucleotide) is a coenzyme - a small helper molecule enzymes need to work - found in every living cell. It has two jobs: carrying energy (electrons) for ATP production, and acting as fuel for control enzymes called sirtuins and PARPs. That dual role ties cellular energy to research on ageing, metabolism and stress.

## What's been studied

- Redox cycling: swaps between “loaded” (NAD<sup>+</sup>) and “unloaded” (NADH) forms; the ratio is read as a quick measure of a cell's energy state.
- Sirtuins consume NAD<sup>+</sup> to do their work, so the cell's NAD<sup>+</sup> supply directly limits sirtuin activity in studies.
- PARP enzymes burn through NAD<sup>+</sup> during DNA repair, creating a tug-of-war with sirtuins when DNA is damaged.
- NAD<sup>+</sup> levels drop with age in lab animals, driving research on precursors (NMN, NR) with NAD<sup>+</sup> as the reference reagent.

**SUPPLIED AS****Lyophilised powder · ≥99.5% HPLC · batch CoA on request.**

CATEGORY

# Recovery

Peptides studied for tissue repair, blood-vessel growth and inflammation.

IN THIS SECTION

- **BPC-157**
- **TB-500**
- **KPV**
- **GHK-Cu**

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# BPC-157

**CAS****137525-51-0****CLASS****Synthetic pentadecapeptide ·  
GEPPPGKPADDAGLV (15 aa)****HPLC****≥99.4%**

## What it is

BPC-157 (Body Protection Compound-157) is a lab-made pentadecapeptide (15 amino acids) copied from a small piece of a protein found in stomach juice. It is studied in animal models for tissue repair, angiogenesis (growth of new blood vessels) and gut protection, and is notably stable in water and stomach-like conditions.

## What's been studied

- Linked to angiogenesis via the VEGFR2 signal and nitric oxide, observed in endothelial cells (which line blood vessels).
- In tendon fibroblasts (cells that build and repair tendons), reported changes in cell growth, survival, movement and growth-hormone receptors.
- Because it derives from a stomach protein, widely studied on the gut lining and the gut-brain axis in injury models.
- Often paired with TB-500 in tissue-repair models, where the two are thought to act in complementary ways.

**SUPPLIED AS****Lyophilised powder · ≥99.4% HPLC · batch CoA on request.**



# TB-500

**CAS****77591-33-4****CLASS****Actin-binding peptide · from  
Thymosin  $\beta$ 4 (43 aa)****HPLC** **$\geq 99\%$** 

## What it is

TB-500 is a lab-made peptide built around the active part of Thymosin  $\beta$ 4, a natural protein found in nearly every cell and abundant in platelets and wound fluid. It is studied for cell migration, angiogenesis and tissue repair. Its core job is managing actin - the protein “scaffolding” cells rebuild in order to move.

## What's been studied

- Holds loose actin pieces (G-actin) in reserve, giving cells a ready supply to build the “feet” that pull them forward in wound healing.
- Linked to blood-vessel cells moving, forming tubes and growing new vessels (angiogenesis); reported to speed skin-wound repair in young and old animals.
- Studied for helping inflammation settle by lowering cytokines and for remodelling the matrix between cells via matrix metalloproteinases.
- The “help cells move and survive” theme also studied in heart and cornea repair models.

**SUPPLIED AS****Lyophilised powder ·  $\geq 99\%$  HPLC (third-party) · batch CoA on request.**

# KPV

## CAS

13147-97-2

## CLASS

Tripeptide · Lys-Pro-Val ·  
C-terminal  $\alpha$ -MSH fragment

## HPLC

 $\geq 98.5\%$ 

## What it is

KPV is a tripeptide (three amino acids: lysine, proline, valine) - the tail-end piece of the hormone alpha-melanocyte-stimulating hormone ( $\alpha$ -MSH). It keeps the calming, anti-inflammatory part of  $\alpha$ -MSH while dropping the colour-changing part, making it a clean research tool for studying inflammation in lab tests.

## What's been studied

- Turns down NF- $\kappa$ B - a master “on switch” for inflammation - and quiets the MAP-kinase pathway, lowering cytokine release even at very low amounts.
- Enters cells through the PepT1 transporter and works without using the melanocortin receptor, unlike full  $\alpha$ -MSH.
- In mouse colitis models, oral KPV reported to reduce severity and lower inflammation messengers.
- Also studied in skin and wound inflammation, often alongside other repair peptides.

## SUPPLIED AS

Lyophilised powder ·  $\geq 98.5\%$  HPLC · batch CoA on request.

# GHK-Cu

**CAS****49557-75-7****CLASS****Copper-binding tripeptide ·  
Gly-His-Lys · Cu(II)****HPLC****≥99%**

## What it is

GHK-Cu (copper-tripeptide-1) is a tripeptide (glycine, histidine, lysine) bound to a single copper ion. First found in human blood plasma in the 1970s, its natural levels fall with age. It is studied as a research tool in tissue repair, skin-cell biology and antioxidant pathways.

## What's been studied

- In dermal fibroblasts (skin support cells), linked to switching on genes for type I collagen, elastin, decorin and glycosaminoglycans.
- Studied as a tool for how copper moves through tissue and for antioxidant enzymes like superoxide dismutase under oxidative stress.
- In rodent and lab-dish wound models, observed alongside angiogenesis, fibroblast movement and rebuilding of the tissue support structure.
- The lab-made version is chemically identical to the natural one, with copper attached.

**SUPPLIED AS****Lyophilised powder · ≥99% HPLC · batch CoA on request.**

CATEGORY

# Growth

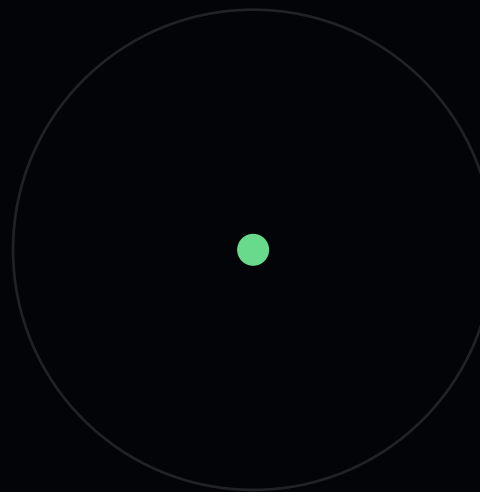
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GHRH analogues studied for prompting the body's own growth-hormone bursts.

IN THIS SECTION

- **CJC-1295**
- **Tesamorelin**

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# CJC-1295

**VARIANT****without DAC · Modified GRF(1-29)****CLASS****GHRH analogue · target GHRH receptor****HPLC****≥99.1%**

## What it is

CJC-1295 without DAC (Modified GRF(1-29)) is a lab-made peptide that acts like GHRH, the natural hormone telling the body to release its own growth hormone in short bursts. Four swapped amino acids protect it from enzymes, so it lasts longer than plain GHRH(1-29) - but, without the DAC handle, only for about half an hour.

## What's been studied

- Docks onto the GHRH receptor on pituitary somatotroph cells (which make growth hormone), raising the cell messenger cAMP.
- Acts one step before growth hormone, so the gland keeps its own natural controls in research models.
- Short ~30-minute action studied for producing separate GH bursts resembling the body's natural rhythm.
- Studies report preserved pulsatility with higher GH and higher IGF-1, rather than a flattened rhythm.

**SUPPLIED AS****Lyophilised powder · ≥99.1% HPLC · batch CoA on request.**

# Tesamorelin

**CAS****218949-48-5****CLASS****GHRH analogue · target GHRH receptor****HPLC****≥99%**

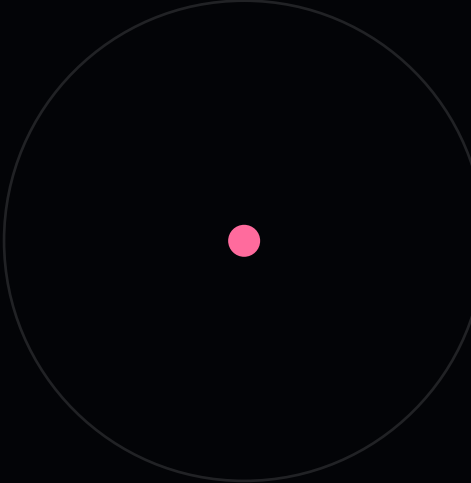
## What it is

Tesamorelin is a GHRH analogue - the full 44-amino-acid GHRH chain with an added cap (a trans-3-hexenoyl group) that acts like armour against enzyme breakdown. It fits the GHRH receptor on the pituitary and is studied as a tool for prompting the body to release its own growth hormone in natural bursts.

## What's been studied

- Binds the GHRH receptor on pituitary somatotrophs, raising cAMP, which makes the cell build and release growth hormone.
- Works one step earlier than giving GH directly, keeping the body's natural burst rhythm and feedback controls intact.
- Released GH then triggers IGF-1, extending research past the pituitary to the wider GH/IGF-1 system.
- Most studied for metabolic effects - especially on visceral fat (deep fat around the organs) and how the body handles fats.

**SUPPLIED AS****Lyophilised powder · ≥99% HPLC · batch CoA on request.**



CATEGORY

# Antioxidant

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Molecules studied for cellular redox, defence and melanocortin signalling.

IN THIS SECTION

- **Glutathione**
- **Melanotan II**

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# Glutathione

**CAS****70-18-8****CLASS****Tripeptide ·  $\gamma$ -Glu-Cys-Gly (GSH)****HPLC** **$\geq 99.2\%$** 

## What it is

Glutathione (GSH) is a tripeptide (glutamate, cysteine, glycine) and the most common antioxidant inside our cells - a molecule that mops up cell-damaging particles. A sulphur-containing thiol group on its cysteine does the work, handing out electrons to neutralise reactive oxygen species. It is a key reference chemical in oxidative-stress research.

## What's been studied

- GSH/GSSG cycle: gives up electrons to become the used-up form (GSSG), then is recycled back to fresh GSH by glutathione reductase using NADPH.
- Fuels glutathione peroxidase (GPx) enzymes that break down hydrogen peroxide and damaged fats into safe products.
- Via glutathione S-transferases, sticks onto unwanted or foreign substances so the cell can clear them (detoxification).
- The GSH-to-GSSG ratio shifts measurably under stress, so it is used as a marker of how stressed a cell is.

**SUPPLIED AS****Lyophilised powder (GSH form) ·  $\geq 99.2\%$  HPLC · batch CoA on request.**



# Melanotan II

**CLASS****Cyclic  $\alpha$ -MSH analogue****TARGETS****MC1R, MC3R, MC4R, MC5R  
(mostly skips MC2R)****HPLC** **$\geq 99.3\%$** 

## What it is

Melanotan II (MT-2) is a lab-made cyclic peptide copied from the hormone  $\alpha$ -MSH. Its ring shape makes it tougher and longer-lasting than the natural hormone. In lab tests it switches on four of the five melanocortin receptors, making it a broad reference tool for studying how those switches work. Not for tanning or cosmetic use of any kind.

## What's been studied

- Non-selective agonist: turns on MC1R, MC3R, MC4R and MC5R at once while mostly skipping MC2R - its main studied feature.
- MC1R sits on melanocytes (pigment-making cells); MT-2 is used to study how this switch controls pigment production.
- MC3R and MC4R are brain/nerve switches studied for energy use and feeding behaviour in rodent models.
- Because it hits several receptors, used as a “yardstick” to benchmark newer, more selective compounds.

**SUPPLIED AS****Lyophilised powder ·  $\geq 99.3\%$  HPLC · batch CoA on request.**

# Glossary

**Agonist**

A molecule that binds a receptor and switches it on.

**Amino acids**

The building blocks that link together to form peptides and proteins.

**Angiogenesis**

The growth of new blood vessels.

**Bacteriostatic water**

Sterile water with 0.9% benzyl alcohol that limits bacterial growth.

**BDNF**

Brain-derived neurotrophic factor; helps nerve cells grow and survive.

**CoA**

Certificate of Analysis - the lab report for a specific batch.

**HPLC**

High-performance liquid chromatography; measures sample purity.

**In vitro**

In glass - test-tube and cell-culture laboratory work.

**Lyophilised**

Freeze-dried into a dry, stable powder.

**Mass spectrometry**

Measures molecular weight to confirm a peptide's identity.

**Peptide**

A short chain of amino acids.

**Receptor**

A cell-surface "lock" a signalling molecule fits into.

**Reconstitution**

Mixing a freeze-dried powder back into a liquid.

**Redox**

Reactions that pass electrons back and forth between molecules.

**Sirtuins**

Enzymes that use NAD<sup>+</sup> as fuel; tied to metabolism and ageing.

# Why Velox

- **≥99% HPLC purity**

Every compound verified by high-performance liquid chromatography.

- **CoA per batch**

A Certificate of Analysis available on request for the exact batch.

- **Free UK shipping over £80**

On qualifying UK orders.

- **Janoshik third-party testing**

Independent lab results - not self-reported numbers.

- **24h Royal Mail Tracked**

UK dispatch within 24 hours, tracked end to end.

- **Pay by Bank**

Fast, secure bank-to-bank checkout.

## How to order

Browse the full range and place an order at [veloxpeps.com/compounds/](https://veloxpeps.com/compounds/). Use the reconstitution calculator at [veloxpeps.com/tools/reconstitution-calculator/](https://veloxpeps.com/tools/reconstitution-calculator/) to plan concentrations before you begin.

**Research-grade peptides, properly documented.**

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